

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2018

Barracks Road 138 kV
March 19, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2018. The Interconnection Customer proposes a 87.5 MW solar plus storage facility interconnecting at the Barracks Road 138 kV substation in the Virginia area. The request was received on February 25, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 677 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,708,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GROVE ST-CHARLOTTESVILLE 138 KV line	COLGATE-ALBRED TANS MTN TAP 138 KV Line	148.0	240	34.7
PENINSULA SUBSTATION-PINE HOLLOW 138 KV	FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV	141.8	240	13.0
GORDONSVILLE-PROFFIT 138 KV	FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV	136.2	364	13.0
PROFFIT-COLGATE 138 KV L	LOUISA GENERATION FACILITY-FLUVANNA P.S 345 KV line	134.2	385	13.0
ALBRED TANS MTN TAP-COLGATE 138 KV	FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV line	130.8	429	13.0
GORDONSVILLE-GORDONSVILLE ENERGY LP SUBSTATION 138 KV	FLUVANNA P.S-LOUISA GENERATION FACILITY SUBSTATION 345 KV	129.6	378	9.9
GORDONSVILLE ENERGY LP-LITTLEFORK 138 KV	LOUISA GENERATION FACILITY-FLUVANNA P.S 345 KV	123.2	496	5.3
PENINSULA-RUNNING Y 138 KV	FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV	122.1	271	13.0
GORDONSVILLE ENERGY LP SUBSTATION-LOUISA GENERATION FACILITY 138 KV Line 1	LOUISA GENERATION FACILITY-FLUVANNA P.S 345 KV	117.8	444	6.7
LOUISA GENERATION FACILITY-GORDONSVILLE ENERGY LP 138 KV Ckt 2	FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV	117.8	444	6.7
ALBRED TANS MTN TAP-RUNNINGY 138 KV	FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV	113.9	282	13.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
AMBER-BREMO BLUFF 138 KV	LAKE LURE-BLUFF CITY 138 KV line	111.0	406	8.8
WALWORTH-BLUFF CITY 138 KV line	FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV L	109.5	474	6.8
GROVE STREET-FLUVANNA P.S 138 KV	RED HILL-FLUVANNA P.S 345 KV Line	108.5	345	28.6
THOMASTON-COMANCHE PEAK 345 KV	THOMASTON-HELIJET 345 KV Line	106.6	1,551	6.1
HELIJET-THOMASTON 345 KV	COMANCHE PEAK-THOMASTON 345 KV	106.6	1,545	5.9
CITY WEST DIESEL PLANT-NESTUCA 345 KV	NAUVOO-CITY W DIESEL PLANT 345 KV	103.6	1,740	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GROVE ST-CHARLOTTESVILLE 138 KV	Rebuild line with higher-capacity conductor	\$12,487,000	17% (\$2,109,000)
PENINSULA SUBSTATION-PINE HOLLOW 138 KV	Rebuild line with higher-capacity conductor	\$3,328,000	10% (\$337,000)
GORDONSVILLE-PROFFIT 138 KV	Rebuild line with higher-capacity conductor	\$11,917,000	8% (\$931,000)
PROFFIT-COLGATE 138 KV L	Rebuild line with higher-capacity conductor	\$8,791,000	8% (\$687,000)
ALBRED A TANS MTN TAP-COLGATE 138 KV	Rebuild line with higher-capacity conductor	\$3,591,000	8% (\$281,000)
GORDONSVILLE-GORDONSVILLE ENERGY LP SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$5,590,000	7% (\$406,000)
GORDONSVILLE ENERGY LP-LITTLEFORK 138 KV	Rebuild line with higher-capacity conductor	\$4,613,000	4% (\$165,000)
PENINSULA-RUNNING Y 138 KV	Rebuild line with higher-capacity conductor	\$11,662,000	10% (\$1,180,000)
GORDONSVILLE ENERGY LP SUBSTATION-LOUISA GENERATION FACILITY 138 KV Line 1	Rebuild line with higher-capacity conductor	\$1,559,000	6% (\$92,000)
GORDONSVILLE ENERGY LP-LOUISA GENERATION FACILITY 138 KV Line 2	Rebuild line with higher-capacity conductor	\$2,028,000	6% (\$120,000)
ALBRED A TANS MTN TAP-RUNNINGY 138 KV	Rebuild line with higher-capacity conductor	\$4,386,000	10% (\$444,000)
AMBER-BREMO BLUFF 138 KV	Rebuild line with higher-capacity conductor	\$18,267,000	5% (\$973,000)
WALWORTH-BLUFF CITY 138 KV	Reconductor line	\$1,587,000	7% (\$108,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GROVE STREET-FLUVANNA P.S 138 KV	Reconductor line	\$668,000	21% (\$139,000)
THOMASTON-COMANCHE PEAK 345 KV L	Reconductor line	\$14,084,000	2% (\$285,000)
HELIJET-THOMASTON 345 KV	Reconductor line	\$13,538,000	2% (\$294,000)
CITY WEST DIESEL PLANT-NESTUCA 345 KV	Reconductor line	\$17,191,000	1% (\$157,000)
	Total allocated to project		\$8,708,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.